

Glossary of Set Notation Symbols

1 Set Operations Glossary

- $\emptyset$: The **empty set**, the set with no elements.

$$\emptyset = \{\}$$

- $\in$: **Element of**. Denotes that an element is part of a set.

$$x \in A \quad \text{means} \quad x \text{ is an element of set } A$$

- $\notin$: **Not an element of**. Denotes that an element is not part of a set.

$$x \notin A \quad \text{means} \quad x \text{ is not an element of set } A$$

- $\subseteq$: **Subset**. Set A is a subset of set B if all elements of A are also elements of B .

$$A \subseteq B \quad \text{means} \quad \text{every element of } A \text{ is also in } B$$

- $\subset$: **Proper subset**. Set A is a proper subset of B if all elements of A are in B , but $A \neq B$.

$$A \subset B \quad \text{means} \quad A \text{ is a proper subset of } B$$

- $\cup$: **Union**. The union of sets A and B is the set of elements that are in A , B , or both.

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

- $\cap$: **Intersection**. The intersection of sets A and B is the set of elements that are in both A and B .

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

- $\setminus$: **Set difference**. The difference between sets A and B is the set of elements that are in A but not in B .

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$$

- $\triangle$: **Symmetric difference**. The symmetric difference of sets A and B is the set of elements that are in either A or B , but not in both.

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

- $\mathbb{U}$: **Universal set**. The set containing all possible elements under consideration in a particular discussion.

$$A \subseteq \mathbb{U} \quad \text{where } \mathbb{U} \text{ is the universal set.}$$

- A^c : **Complement**. The complement of set A is the set of all elements in the universal set $\mathbb{U}$ that are not in A .

$$A^c = \{x \mid x \in \mathbb{U} \text{ and } x \notin A\}$$

- $|A|$: **Cardinality**. The cardinality of a set is the number of elements in the set A .

$$|A| = \text{number of elements in } A$$